

Gated Progressive Formalization: A Method for Disciplining the Conversion of Messy Conceptual Material into Better-Constrained Models

Complete Working Paper — Final Consolidated Version

Includes core method, gap-directed search, supporting procedures,
two live applications, and the under-determination vs. falsehood distinction

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Abstract

This paper presents a complete methodological system for converting messy, metaphor-rich, or advocacy-laden conceptual material into clearer, better-constrained models under explicit discipline. The core mechanism is a gated progressive process: new formal layers are admitted only when they satisfy conditions of consistency with anchors, inter-subjective agreement, productive yield of checkable consequences, and bounded residual ambiguity, plus higher-level review. A revision path is required. When gates fail, a gap-directed search loop converts the failure into a concrete research agenda; candidate material must itself clear a relevance-plus-consistency check before incorporation. Supporting procedures further discipline higher-order questions of relevance and importance, including gap ranking, search-completeness rules, anchor stress-testing, comparative direction evaluation, and classical sound-argument training for relevance. Residual evaluative judgment is catalogued explicitly. Two live applications are reported: (1) a computational sketch of conscience, which was judged admissible but provisional after controlled tightening; and (2) a contemporary opinion article on the rise of democratic socialism, which underwent two full search-and-re-score cycles. The second application demonstrates that the method can progressively constrain factual sub-claims while continuing to block elevation of stronger interpretive claims that remain under-determined. High residual ambiguity after serious search does not prove a claim false; it establishes that the claim has not been elevated to a well-constrained, intersubjectively stable status and therefore remains provisional or theoretical relative to the accepted anchors. The method functions as a diagnostic that surfaces shortcomings and generates an actionable agenda for progress toward better-constrained formulations. It does not eliminate judgment, prove relevance in a strong sense, or supply a general algorithm for insight.

1. Introduction and Problem Statement

Conceptual conversations, informal theories, and advocacy arguments frequently contain suggestive metaphors, partial insights, causal claims, and implicit structure. Turning that material into clearer, usable, better-constrained models is a recurring intellectual task. No efficient general algorithm currently exists for doing so reliably. The central difficulty is the absence of a tractable success signal for intellectual depth, appropriate abstraction, and non-obvious capture of structure. Scaling compute or data does not dissolve the problem when the target qualities themselves remain poorly specified.

This paper develops a practical response: a gated progressive formalization process that makes under-determination visible, constrains unconstrained elaboration, forces residual judgment into the open, and converts identified gaps into a directed research agenda. The method is designed to be honest about its limits. It localizes and scaffolds evaluative judgment; it does not claim to eliminate it. Two live applications illustrate both the diagnostic power of the process and the distinction between claims that are under-determined and claims that have been shown false.

2. The Core Gated Process

L_0 = objective anchors (empirical results, logical constraints, strong intersubjective facts)

L_{k+1} is admitted only if all of the following hold:

1. $\text{Cons}(L_{k+1}, L_0 \text{ union prior layers}) \geq \tau_c$
2. $\text{Agree}(L_{k+1}) \geq \tau_a$
3. $\text{Prod}(L_{k+1}) \geq \tau_p$
4. $\text{Amb}(L_{k+1}) \leq \tau_u$
5. Higher-level review against a fixed rubric passes

Revision rule: Any earlier layer may be updated or withdrawn if inconsistency appears. Revisions must re-satisfy the gates.

Required annotations: constrained claims; free/under-determined claims;
reliability estimate; any revisions forced.

2.1 Operational Gates

Cons — compatibility rate with anchors and prior layers. Concrete contradictions, failed predictions, or logical inconsistencies count against it.

Agree — concordance across careful reviewers or models on a written rubric.

Prod — number of new non-trivial, checkable consequences generated by the layer.

Amb — remaining free parameters or alternative readings still equally compatible with current anchors. High Amb blocks expansion.

Higher-level review — fixed rubric checking overclaim, category error, and silent meaning shifts.

3. Gap-Directed Search Loop

When a gate fails (especially Amb), the process converts the failure into a directed research step rather than permitting unconstrained further elaboration:

1. Extract the specific free parameters or missing constraints responsible.
2. Actively search for new material that targets those exact gaps.
3. Evaluate candidate material for relevance to the targeted gap and consistency with existing anchors/layers. Only material that clears this check is admissible for incorporation.
4. Incorporate clearing material and re-score the gates.
5. Repeat until the gates clear or the search returns nothing that meaningfully reduces the failure — then remain provisional and stop expanding the formal structure.

This preserves the hard pause on unconstrained elaboration while turning identified shortcomings into an actionable agenda. Expansion resumes only after gaps have been constrained and the gates re-pass.

4. Supporting Procedures for Higher-Order Relevance and Importance

Deeper questions (importance of a gap, existence of better material, completeness of anchors, overall significance versus local coherence) cannot be fully decided by the core gates. Supporting procedures make them more disciplined:

Gap ranking heuristics — score gaps by number of other parameters affected, directness of anchor connection, and estimated measurability.

Search-completeness rules — declare source classes and a diminishing-returns or time-box stopping rule; record what was checked.

Anchor stress-testing — seek potential defeating evidence or alternative anchor sets and re-score under them.

Comparative direction evaluation — run alternative formalization paths under the same gates and compare residual Amb, yield, and stability.

Classical sound-argument training for relevance — use historically successful, widely accepted arguments as positive exemplars of material that closed important gaps; condition a relevance heuristic on the resulting patterns. Limits: exemplar selection is itself judgment; relevance is context-dependent; past success does not guarantee performance on novel gaps.

5. Residual Judgment — Explicit Catalogue

The method scaffolds judgment and makes its consequences more detectable. It does not remove it. Principal residual loci include:

- Choice and weighting of initial anchors.
- Calibration of numerical thresholds.
- Deciding that particular material is relevant to a specific gap.
- Ranking gaps and declaring a search sufficient.
- Assessing overall significance versus mere local coherence.
- Selecting classical arguments as sound exemplars for training.
- Interpreting borderline cases and deciding that a consequence is non-trivial.

Naming these points keeps the process honest. Overclaiming that residual judgment has been eliminated violates the method's own conditions.

6. First Live Application: Computational Sketch of Conscience

Informal metaphors of conscience (recursive loop, genetic seed, partial privacy/writability, relational growth) were formalized into a computational sketch. Four modest developmental anchors were locked. The innate seed G was tightened to a minimal three-component structure (harm, fairness, social sensitivity). After re-scoring the layer was judged **admissible but provisional**. Ambiguity on functional forms and the dynamics of self-relationship remained the binding constraint. The method permitted one controlled improvement and then correctly blocked further unconstrained expansion.

7. Second Live Application: Opinion Article on Democratic Socialism

An August 2026 opinion article arguing that socialism is moving into the American mainstream (centered on NYC Mayor Zohran Mamdani and the DSA), that specific tax and transparency actions constitute intimidation and a transferable blueprint, and that the correct response is public exposure and labeling the actors communists, was run through the full process.

7.1 Initial Diagnostic

Modest anchors were locked (existence and activity of DSA; documented policies of the mayor; possibility of policy diffusion; historical record of many socialist/communist regimes; influence of public framing). The article's stronger claims were treated as a candidate layer. Cons was only moderate; Prod was weak-to-moderate; Amb was high and binding. Verdict: not admissible as a tightly constrained formal layer.

7.2 Gap-Directed Searches and Re-scoring

Two full search cycles were executed. Candidate material was required to clear a relevance-plus-consistency check before incorporation. Admitted material constrained: (a) measurable electoral growth of the Mamdani/DSA orbit inside major progressive cities, especially NYC primaries; (b) the expansive public character of the property-tax database rollout paired with celebratory targeting of wealthy non-primary owners; (c) geographic concentration of success in deep-blue urban areas and the still-limited translation beyond them; (d) documented radical platform content alongside contested public branding and internal ideological diversity on the "communist" label. After incorporation, Cons remained high, factual Agree improved, Prod gained modestly, and Amb was reduced on the targeted gaps but remained elevated on the strongest original claims (full mainstream capture, inevitable national spread, necessity of the communist label).

7.3 Final Status of the Article Application

The core warning (growing organized socialist electoral power centered on NYC/DSA, using aggressive wealth-targeting tactics, with intent to expand) became better constrained and can be treated as a supported provisional claim. The stronger assertions of already-achieved mainstream capture, imminent national diffusion, and the necessity of the communist label remained under-determined and did not clear Amb. The method progressively tightened the factual base while continuing to block full elevation of the most expansive claims.

8. Under-determination versus Demonstrated Falsehood

A central clarification produced by the second application concerns the proper reading of persistent high Amb. High residual ambiguity after serious gap-directed search does **not** prove a claim false. It establishes that the claim has not been elevated to a well-constrained, intersubjectively stable status relative to the accepted anchors. The claim remains provisional, interpretive, or theoretical.

Clear signs of a false claim appear only when the process produces a hard, stable Cons failure: direct incompatibility with anchors or admitted material that survives scrutiny, converges across independent sources, produces predictive or observational failure, cannot be restored by legitimate revision, and forces alternative readings to collapse. The method deliberately keeps "not yet established" and "demonstrated false" as distinct outcomes.

Thus the process can show that a claim is not objective (not well-constrained) without thereby proving it false. This distinction prevents both over-elevation of under-supported claims and premature dismissal of claims that are still open.

9. What the Method Accomplishes

Applied consistently, the method surfaces specific shortcomings — free parameters, under-determined claims, and places where a formulation still floats free of its anchors. It converts those shortcomings into a concrete list of what must be constrained if the idea is to move closer to a well-supported status. It blocks the common habit of treating under-determined claims as if they were already established. It functions as a progress diagnostic and a research-agenda generator rather than as a final verdict machine.

It can be applied to informal theories, advocacy arguments, policy claims, and other forms of messy conceptual material. In each case the same logic holds: make the distance from the current formulation to a better-constrained one visible and actionable, and refuse to let unconstrained elaboration paper over the gaps.

10. Limits and Honesty Conditions

The method does not eliminate evaluative judgment, prove relevance in a strong external sense, guarantee that the most important gap or best material has been found, or supply a general algorithm for insight. Residual judgment remains at the points catalogued above. Treating the method as having removed that judgment, or as having converted high Amb directly into falsehood, would itself violate the honesty conditions the process is designed to enforce.

11. Conclusion

Gated Progressive Formalization, extended with a gap-directed search loop that itself subjects new material to admission checks, together with supporting procedures for higher-order relevance questions, is offered as the strongest realistic methodological resolution currently available for disciplining the conversion of messy conceptual material into better-constrained models. Two live applications confirm that the process can enforce controlled improvement, surface specific shortcomings, generate an actionable research agenda, and maintain a principled distinction between under-determination and demonstrated falsehood. Further value will be determined by additional applications, refinement of the operational measures, and continued refusal to overclaim what residual judgment must still supply.

Final consolidated working paper — August 2026. This document incorporates the complete current formulation of the method, both live applications, the under-determination vs. falsehood distinction, and the method's function as a diagnostic and research-agenda generator. It is a methodological proposal offered for scrutiny and use, not a finished theoretical system.